



$$F_{CFU} = 15397 \times \frac{V(V)}{V_{ref}}$$

(3) RMS-current output pulse frequency:

$$F_{CFI} = 94638 \times \frac{V(I)}{V_{ref}}$$

V(V) — RMS voltage at the voltage-channel input

V(I) — RMS voltage at the current-channel input

Vref — Reference voltage (typical: 1.1V)

Note: The above formulas are based on typical values.

8.3 Anti-Creep

The BL0937B features a patented anti-creep design. When used with an appropriate external hardware design, it ensures that power due to noise is not accumulated as energy pulses when no current is present. The anti-creep threshold is 3.5/100,000 of the active power corresponding to a full-scale input signal.

8.4 Overcurrent Detection

The BL0937B incorporates fast overcurrent detection, which detects a current overload within 200ms and simultaneously outputs an overcurrent indication signal at the CF pin. This facilitates the design of overcurrent-protection circuits.

8.5 RMS Current/Voltage Output

The BL0937B outputs the SEL-selected RMS current or voltage signal at the CF1 pin. When SEL=0, the CF1 pin outputs a high-frequency pulse train whose frequency corresponds to the RMS current; when SEL=1, the CF1 pin outputs a high-frequency pulse train whose frequency corresponds to the RMS voltage. The internal RMS-current and RMS-voltage calculation modules operate independently, and the SEL switching latency is less than 10 μs.